



Efficient Streaming TTS Acoustic Model with Depthwise RVQ Decoding Strategies in a Mamba Framework

*Joun Yeop Lee, Sangjun Park, Byoung Jin Choi, Ji-Hyun Lee,
Min-Kyung Kim, Hoon-Young Cho*

Samsung Research, Samsung Electronics, Republic of Korea

{jounyeop.lee, sj0.park, bj87.choi, jihyuny.lee, mk_0.kim, h.y.cho}@samsung.com

Abstract

Recent advances in neural codec-based text-to-speech (TTS) systems have achieved remarkable synthesized speech quality. However, their reliance on large model size and heavy computational requirements limits CPU-based on-device deployment. In this work, we present a Mamba-based streaming acoustic model with two novel depthwise decoding strategies for residual vector quantization (RVQ)-based codec: a Masked Language Model (MLM) approach and an Implicit Neural Representation (INR) approach. The MLM strategy iteratively refines tokens along the code depth axis to enhance speech quality, whereas the INR approach predicts all quantization levels in parallel to reduce computational costs. We further incorporate a speaker embedding conditioning mechanism for a zero-shot scenario, enabling robust performance on unseen speakers. Experimental results demonstrate comparable or even superior improvements in both objective and subjective metrics compared to other larger TTS baseline models.

Index Terms: streaming speech synthesis, Mamba, depthwise decoding, masked language model, implicit neural representation

1. Introduction

Recent advancements in text-to-speech (TTS) technology have seen a paradigm shift from traditional continuous intermediate features, such as mel-spectrogram [1, 2, 3], to discrete token-based intermediate features [4, 5, 6, 7]. This shift has resulted in significant performance improvements, driven largely by large-scale models and language model-inspired methodologies. These token-based approaches have enabled more fine-grained acoustic modeling and better integration with language models, contributing to remarkable gains in speech synthesis quality.

However, while these advancements excel in high-resource, server-based environments, they do not inherently address the long-standing challenges of computational cost and real-time performance, which are critical for on-device applications. The computational demands of large models remain a bottleneck when deploying TTS systems on resource-constrained hardware, where latency and efficiency are important. Moreover, a substantial number of current devices have not yet fully exploited the capabilities of parallel computation facilitated by GPU/NPU. Previous works using continuous intermediate feature have adapted a lightweight, streaming acoustic model decoder architecture which has low latency [8]. Similarly, we propose a Mamba-based [9] streaming decoder in the acoustic model to bridge this gap for the discrete token-based TTS.

Numerous neural codec models utilize residual vector quantization (RVQ) for speech synthesis [10, 11, 12, 13]. In

these models, multiple quantization tokens must be predicted at each time step. Previous works, such as MusicGen [14] and Small-E [15], have employed delayed decoding strategies that buffer multiple frames before synthesis, leading to increased latency. In contrast, our acoustic model directly predicts RVQ tokens corresponding to each time step, enabling low-latency inference. Our approach introduces two novel depthwise decoding strategies tailored for RVQ:

- Masked Language Model (MLM) decoding strategy that iteratively refines tokens along the code depth axis to capture inter-level dependencies, which enhances synthesis quality.
- Implicit neural representation (INR) strategy that predicts multiple quantization levels simultaneously to reduce computational overhead and inference latency.

These strategies are designed to maintain the benefits of token-based TTS systems while overcoming the computational and latency limitations inherent to on-device deployment.

The MLM approach proposed in this work differs from prior non-autoregressive (NAR) models, such as [16], which consider both the time and depth axes of tokens for masking scheduling. Although effective for NAR generation, such approaches are not well suited for streaming applications. In contrast, our method only considers the depth axis of tokens, ensuring compatibility with streaming inference while still modeling dependencies between tokens.

Unlike prior work where INR has been applied in TTS primarily in alignment modeling, such as in SegINR [17], our approach leverages INR for depthwise prediction. This allows for efficient modeling across quantization levels without requiring additional blank tokens, resulting in a simple and fast depthwise prediction mechanism.

By integrating decoding strategies into a Mamba-based streaming acoustic model, we present a practical solution that balances synthesis quality with the efficiency required for real-time, on-device TTS applications. The following sections provide detailed insights into the core techniques and experimental results. Experimental results on the zero-shot scenario demonstrate that our proposed model achieves comparable or superior performance in terms of speech quality and speaker similarity, despite having lower computational requirements and a smaller model size compared to baseline models. Audio samples are available on our demo page¹.

2. Background

This section provides an overview of the key components that form the foundation of our proposed method: the Mamba framework for efficient sequential modeling, the Masked Lan-

¹<https://srtts.github.io/mamba-depthwise>

guage Model (MLM), and Implicit Neural Representations (INR) for depthwise decoding.

2.1. Mamba

Mamba [9] is a structured state-space model that is specifically designed for efficient sequential data modeling. Unlike Transformer-based architectures, which rely on global self-attention, Mamba processes inputs incrementally. The key advantages of Mamba include:

- **Streaming-Friendly Inference:** It generates outputs incrementally without requiring access to the entire input sequence.
- **Scalability:** Its computational efficiency enables efficient modeling of long sequences.

These properties establish Mamba as an effective alternative to attention-based models in streaming TTS scenarios where real-time performance and resource efficiency are paramount.

2.2. Masked Language Model (MLM)

The MLM [18] has proven its effectiveness in capturing contextual dependencies in natural language processing by training models to predict randomly masked tokens. Inspired by its success in natural language processing, MLM has been adapted to TTS systems [16].

In [16], masking is applied mainly along the time axis during training. Specifically, parts of the token representation sequence are masked randomly, and the model is trained to predict the missing tokens using the surrounding context. This masking strategy enables the model to learn temporal relationships between consecutive tokens, which are crucial for accurate speech synthesis.

During inference, the entire sequence of RVQ representations is initially masked, and tokens are predicted iteratively. MLM’s capability to capture temporal dependencies across masked tokens makes it suitable for generating acoustic representations in high fidelity.

2.3. Implicit Neural Representation (INR)

INRs [19] are a framework for encoding continuous signals, such as audio or images, as continuous functions of their input coordinates. INRs typically use multi-layer perceptrons (MLPs) to learn these mappings, representing complex structures compactly and efficiently without relying on discrete grids.

A key advantage of INRs lies in their ability to encode high-dimensional data as smooth, parameterized functions. Instead of explicitly storing data points or features, INRs model signals as mappings from coordinates (e.g., spatial or temporal indices) to values (e.g., pixel intensities or tokens). This approach allows INRs to generalize well to unseen input resolutions and enables continuous interpolation between points. Especially in this paper, we modify SegINR [17] for depthwise decoding.

This foundational technology provides a basis for efficient, compact representations in various domains, including TTS systems where multiple depth levels of tokens need to be predicted effectively.

3. Proposed Method

In this section, we describe our proposed streaming acoustic model framework built with Mamba for efficient sequential modeling, alongside two novel depthwise decoding strategies for RVQ codec.

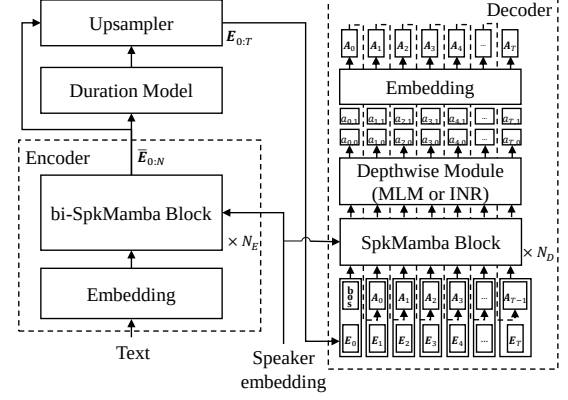


Figure 1: The overall architecture of our proposed acoustic model. Assume depth level of RVQ as two.

3.1. Mamba-based Streaming Acoustic Model

For the streaming acoustic model, as in the typical streaming TTS systems, the entire text sequence $\mathbf{x}_{0:N}$ is given, and acoustic tokens must be generated one at a time to ensure low latency. To achieve this, we design the encoder and streaming decoder utilizing the Mamba-based framework with specific roles and configurations. The overall architecture of the proposed Mamba-based acoustic model is represented in Figure 1.

Building upon the Mamba, we further extend it with speaker conditioning (e.g. SpkMamba) to enable zero-shot speaker adaptation. Specifically, the speaker embedding is injected into the input projection layers of the Mamba block. For speaker representation, we utilize an ECAPA-TDNN-based speaker encoder [20], modified to take acoustic tokens as an input instead of mel spectrograms. This speaker encoder is trained jointly with the acoustic model, ensuring that the learned embeddings are well aligned with the TTS objective.

The encoder processes the entire input text phoneme sequence using a bidirectional Mamba architecture [21] replaced with SpkMamba, enabling the model to leverage the full-text sequence context for accurate text-to-acoustic token mapping. Duration is estimated through the monotonic alignment search during training and a duration predictor at the inference phase. Subsequently, Gaussian upsampling aligns the encoder output sequence with the acoustic token sequence using duration.

The decoder operates autoregressively frame-by-frame. At each timestep t , it concatenates the previous acoustic embedding $\mathbf{A}_{t-1}$, obtained by summing the RVQ token embeddings $\mathbf{a}_{0:D-1,t-1}$ across all D depth levels, with the current upsampled encoder representation $\mathbf{E}_t$. This joint representation is then used to predict the current acoustic token $\mathbf{a}_{0:D-1,t}$ using a depthwise decoding module.

3.2. Depthwise Decoding Strategies for RVQ

To use RVQ codec in TTS, we should predict multiple depth levels of tokens at each acoustic frame. Thus, accurate and efficient decoding of RVQ tokens is critical for high-quality TTS synthesis. Our framework addresses this challenge by introducing two depthwise decoding strategies that operate along the code depth axis. Moreover, these strategies can be applied to various types of TTS systems utilizing RVQ-based neural codecs, including Speech LLM scenarios where the input sequence is also processed in a streaming manner.

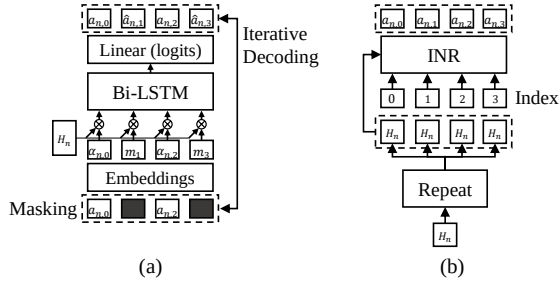


Figure 2: Architectures of (a) MLM-based decoding and (b) INR-based decoding method assuming $D = 4$. H_n represents the output hidden representation of the last SpkMamba block at time n . The sampling procedure is omitted for simplicity.

3.2.1. Masked Language Model (MLM) Decoding

The MLM decoding strategy repurposes the concept of masking from natural language processing. Unlike previous works that mask tokens over the time axis, we focus solely on the depth dimension of RVQ. During training, we employ depth-wise masking, where selected portions of the RVQ representations are masked, and the model is trained to predict these missing values. This strategy encourages the model to learn inter-level dependencies.

At inference time, for a given frame, all depth tokens are initially masked. Then the system iteratively predicts missing tokens following a fixed schedule as in Figure 2(a), applying top- k sampling (with $k = 5$) to maintain diversity. While this iterative procedure increases computation relative to a single-pass approach, it results in high-quality token generation.

3.2.2. Implicit Neural Representation (INR) Decoding

In contrast to the iterative MLM strategy, the INR-based approach leverages depth indices to predict all RVQ quantization levels in parallel as in Figure 2(b). We modify the SegINR model [17], to take input the sequence of depth indices (from 0 to $D - 1$) along with a repeated hidden representation of the last SpkMamba block H_n . By treating the depth indices as continuous coordinates, the INR model generates the complete set of depthwise tokens in a single forward pass, eliminating the need for iterative refinement and explicit blank tokens.

Since INR performs a single forward pass over the depth levels at each frame, it runs faster than MLM-based decoding. However, it may capture slightly fewer cross-depth dependencies, potentially leading to marginally lower synthesis quality compared to MLM. Thus, the INR approach is particularly appealing for resource-constrained applications where minimal latency is a primary goal.

3.3. Integration and Trade-offs

By integrating the proposed acoustic model with our two depth-wise decoding strategies, we offer a flexible solution tailored to diverse deployment scenarios. The MLM-based approach excels in synthesizing high-quality, natural-sounding speech by capturing inter-RVQ level correlations. However, the INR-based method prioritizes speed and computational efficiency by generating tokens in parallel. Consequently, system designers can choose the appropriate decoding strategy based on specific quality or latency requirements, thus enabling effective on-device TTS across a range of hardware specifications.

4. Experiments

In this section, we detail the experimental evaluations of our proposed streaming TTS system, demonstrating the effectiveness and efficiency of our model in a zero-shot scenario.

4.1. Experimental Setup

We conducted experiments on the LibriTTS dataset [22], which consists of multi-speaker recordings. Our training data included the train-clean-100, train-clean-360, and train-other-500 subsets. For evaluation, we selected 500 utterances from the test-clean subset, ensuring a strict zero-shot condition by using unseen speakers. The objective evaluations were performed on all 500 utterances, while subjective evaluations were conducted using a sampled subset consisting of 30 utterances to ensure reliable human assessments.

4.2. Evaluation Metrics

Objective tests were performed using the following metrics:

- **Character Error Rate (CER, %)**: Measures pronunciation accuracy by comparing ASR transcriptions (obtained via a Whisper large model [23]) with ground truth text.
- **Speaker Embedding Cosine Similarity (SECS)**: Quantifies speaker similarity by computing the cosine similarity between embeddings extracted with Resemblyzer².
- **UTMOS**: Predicts perceptual quality via a neural MOS prediction network [24].
- **Real-Time Factor (RTF)**: Assesses computational efficiency by measuring average synthesis time divided by the audio duration across the test set.
- **latency (sec)**: Measures the average time taken for the model to generate the first acoustic token after receiving the input text.

For RTF and latency, we focused solely on the token generation process, excluding codec-related processing, as our method is related to the acoustic model. Both metrics were measured on a single-threaded CPU (AMD EPYC 7413).

Subjective evaluations were conducted through human listening tests on a scale from 1 to 5. The following metrics were used:

- **Mean Opinion Score (MOS)**: Rates overall speech quality including naturalness.
- **Similarity MOS (SMOS)**: Evaluates perceptual speaker similarity compared to a reference sample.

For both subjective tests, 150 testers participated via Amazon MTurk³ platform.

4.3. Implementation Details

For the neural codec, we utilized the official pretrained HiFi-Codec [10] model, which employed a grouped-RVQ approach. The codec consisted of four codebooks, divided into two groups, each containing two codebooks. The model was trained on 24 kHz speech samples, down-sampled by a factor of 320.

For text input, we employed the International Phonetic Alphabet-based phonemizer⁴ to ensure consistent and accurate phonetic representations. Our acoustic model consisted of four bi-SpkMamba blocks (hidden dimension 384) in the encoder

²<https://github.com/resemble-ai/Resemblyzer>

³<https://www.mturk.com>

⁴<https://github.com/bootphon/phonemizer>

Table 1: Comparison of Experimental Results. MOS and SMOS are represented with 95% confidence intervals. Since RTF and latency in this experiment are focused on token generation, VITS is omitted.

	CER	SECS	Parameter	UTMOS	RTF	latency	MOS	SMOS	Streaming
SMAM+MLM	2.73	0.816	26M	4.134	0.701	0.065	4.02 ± 0.11	3.36 ± 0.17	O
SMAM+INR	3.51	0.790	25M	3.920	0.568	0.061	3.97 ± 0.11	3.18 ± 0.17	O
SMAM+noMLM	4.12	0.765	26M	3.829	0.582	0.062	3.89 ± 0.12	3.03 ± 0.17	O
Lee et al. (2024)	4.01	0.863	263M	4.107	4.914	26.53	4.00 ± 0.12	3.27 ± 0.15	X
VALLE-X	6.50	0.846	371M	3.656	7.665	41.41	3.66 ± 0.13	3.12 ± 0.16	X
Small-E	19.65	0.766	63M	3.478	2.404	14.34	3.59 ± 0.15	2.69 ± 0.17	X
VITS	6.52	0.826	46M	3.552	-	-	3.57 ± 0.12	3.04 ± 0.16	X
GT	2.33	0.835	-	4.146	-	-	3.95 ± 0.12	3.89 ± 0.11	-

and eight SpkMamba blocks (hidden dimension 512) in the decoder. The dimension of the embedding layer was 256 in the encoder and 128 in the decoder. The duration predictor was constructed using two bi-LSTM layers, each with 128 dimensions, followed by the projection layer output of one dimension.

The MLM-based decoding approach followed a groupwise masking strategy, as in [16], with three iterations to progressively refine predictions. Also, as in Figure 2 (a), a single bi-LSTM layer with 128 dimensions and 128 embeddings was used for the MLM-based depthwise module. The INR-based depthwise module followed the configuration in [17]. Our proposed model was trained for 600k iterations with a batch size of 64 on a single NVIDIA A100 GPU.

4.4. Comparison Models

The proposed models were compared against state-of-the-art baselines, including VITS [25], VALLE-X [26], the model proposed by Lee et al. (2024) [6], and Small-E [15]. To adapt the baseline VITS for zero-shot, we incorporated the ECAPA-TDNN as the reference encoder. Except for VITS, all three models are token-based TTS systems. Specifically, we used mamba-based Small-E to compare mamba-based structures.

Three configurations of the proposed streaming Mamba acoustic model (SMAM) were evaluated: SMAM with MLM decoding (SMAM+MLM), SMAM with INR decoding (SMAM+INR), and SMAM without advanced decoding strategies (SMAM+noMLM). The SMAM+noMLM, utilized for the ablation study, employs a single-step decoding process without any masking during both the training and inference phases.

All results of our experiments are presented in Table 1.

4.5. Objective Results Analysis

Our experiments demonstrate that, despite having significantly fewer parameters (25–26M) compared to baselines (ranging from 46M to 371M), the proposed methods achieved notable improvements in terms of intelligibility. SMAM+MLM attained a CER of 2.73%, outperforming all the baselines, while SMAM+INR achieved a CER of 3.51%, which is slightly lower than those of SMAM+MLM.

In terms of SECS, although the proposed models did not surpass other baselines, they achieved scores close to the ground truth (GT) benchmark, indicating that speaker similarity is well preserved despite the small model size. The UTMOS results confirmed high perceptual quality, with SMAM+MLM scoring 4.13 compared to lower scores from the larger baseline models.

From a computational perspective, SMAM models demonstrated superior efficiency. SMAM+MLM achieves an RTF of 0.701, while SMAM+INR lowers it further to 0.568, sig-

nificantly outperforming non-streaming baselines, which range from 2.404 to 7.665. By supporting streaming on a CPU, our models maintain latencies below 0.07 seconds. In contrast, baseline models generate entire utterances at once, resulting in latencies of at least 14 seconds, which increase further for longer texts. This substantial improvement emphasizes SMAM’s effectiveness for streaming TTS on CPU devices.

Also, objective results highlight a clear trade-off as mentioned in Section 3.3: while the SMAM+MLM offered the highest synthesis quality, SMAM+INR provided advantages in computational speed and model compactness.

4.6. Subjective Results Analysis

Similar to the UTMOS results, the MOS indicated that the proposed methods deliver better perceptual speech quality, with SMAM+MLM showing the best score of 4.02. The SMOS results differ from those of SECS, as SMAM+MLM exhibited the highest perceptual similarity, scoring 3.36, compared to Lee et al., scoring 3.27. This discrepancy arose since SECS measures embedding space distances, which may not fully align with perceptual similarity judgments. Thus, while SECS reflects overall similarity in the embedding space, SMOS captures subjective perceptions of speaker similarity more accurately. Also, the performance between SMAM+MLM and SMAM+INR supported the trade-off as in the objective results.

4.7. Ablation Study

To show the impact of our depthwise iterative refinement strategy, we compared SMAM+MLM with SMAM+noMLM. The removal of MLM resulted in a severe degradation of all metrics in both objective and subjective tests, except RTF and latency results. These ablation results showed the importance of capturing inter-level dependencies for high-quality synthesis.

5. Conclusion

In this paper, we introduced a zero-shot streaming TTS system based on the Mamba framework (SMAM), incorporating two novel depthwise decoding strategies for RVQ: Masked Language Model (MLM) and Implicit Neural Representation (INR). Our experiments demonstrated that SMAM achieves high computational efficiency while maintaining superior performance compared to larger baseline models, with a clear trade-off between quality (MLM) and speed (INR). These findings enable flexible deployment across diverse hardware constraints, making SMAM a strong candidate for real-time streaming TTS on CPU. Future work will explore fully streaming models across the entire TTS pipeline including codec and apply depthwise decoding strategies to decoder-only speech LLMs.

6. References

- [1] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao, and T.-Y. Liu, “FastSpeech 2: Fast and high-quality end-to-end text to speech,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://openreview.net/forum?id=piLPYqxtWuA>
- [2] J. Y. Lee, J.-S. Bae, S. Mun, J. Lee, J.-H. Lee, H.-Y. Cho, and C. Kim, “Hierarchical timbre-cadence speaker encoder for zero-shot speech synthesis,” in *Interspeech 2023*, 2023, pp. 4334–4338.
- [3] Y. Liu, R. Xue, L. He, X. Tan, and S. Zhao, “Delightfults 2: End-to-end speech synthesis with adversarial vector-quantized auto-encoders,” in *Interspeech 2022*, 2022, pp. 1581–1585.
- [4] E. Kharitonov, D. Vincent, Z. Borsos, R. Marinier, S. Girgin, O. Pietquin, M. Sharifi, M. Tagliasacchi, and N. Zeghidour, “Speak, read and prompt: High-fidelity text-to-speech with minimal supervision,” *Transactions of the Association for Computational Linguistics*, vol. 11, pp. 1703–1718, 2023.
- [5] C. Wang, S. Chen, Y. Wu, Z. Zhang, L. Zhou, S. Liu, Z. Chen, Y. Liu, H. Wang, J. Li *et al.*, “Neural codec language models are zero-shot text to speech synthesizers,” *arXiv preprint arXiv:2301.02111*, 2023.
- [6] J. Y. Lee, M. Jeong, M. Kim, J.-H. Lee, H.-Y. Cho, and N. S. Kim, “High fidelity text-to-speech via discrete tokens using token transducer and group masked language model,” in *Interspeech 2024*, 2024, pp. 3445–3449.
- [7] M. Łajszczak, G. Cámbara, Y. Li, F. Beyhan, A. van Korlaar, F. Yang, A. Joly, Álvaro Martín-Cortinas, A. Abbas, A. Michalski, A. Moinet, S. Karlapati, E. Muszyńska, H. Guo, B. Putrycz, S. L. Gambino, K. Yoo, E. Sokolova, and T. Drugman, “Base tts: Lessons from building a billion-parameter text-to-speech model on 100k hours of data,” 2024. [Online]. Available: <https://arxiv.org/abs/2402.08093>
- [8] N. Ellinas, G. Vamvoukakis, K. Markopoulos, A. Chalamandaris, G. Maniati, P. Kakoulidis, S. Raptis, J. S. Sung, H. Park, and P. Tsiakoulis, “High quality streaming speech synthesis with low, sentence-length-independent latency,” in *Interspeech 2020*, 2020, pp. 2022–2026.
- [9] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” 2024. [Online]. Available: <https://arxiv.org/abs/2312.00752>
- [10] D. Yang, S. Liu, R. Huang, J. Tian, C. Weng, and Y. Zou, “Hifi-codec: Group-residual vector quantization for high fidelity audio codec,” *arXiv preprint arXiv:2305.02765*, 2023.
- [11] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, “Soundstream: An end-to-end neural audio codec,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 495–507, 2021.
- [12] A. Défossez, J. Copet, G. Synnaeve, and Y. Adi, “High fidelity neural audio compression,” *Transactions on Machine Learning Research*, 2023, featured Certification, Reproducibility Certification.
- [13] Z. Ye, P. Sun, J. Lei, H. Lin, X. Tan, Z. Dai, Q. Kong, J. Chen, J. Pan, Q. Liu, Y. Guo, and W. Xue, “Codec does matter: Exploring the semantic shortcoming of codec for audio language model,” *arXiv preprint arXiv:2408.17175*, 2024.
- [14] J. Copet, F. Kreuk, I. Gat, T. Remez, D. Kant, G. Synnaeve, Y. Adi, and A. Defossez, “Simple and controllable music generation,” in *Advances in Neural Information Processing Systems*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., vol. 36. Curran Associates, Inc., 2023, pp. 47 704–47 720. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/file/94b472a1842cd7c56dcb125fb2765fbd-Paper-Conference.pdf
- [15] T. Lemerle, N. Obin, and A. Roebel, “Small-e: Small language model with linear attention for efficient speech synthesis,” in *Interspeech 2024*, 2024, pp. 3420–3424.
- [16] M. Jeong, M. Kim, J. Y. Lee, and N. S. Kim, “Efficient parallel audio generation using group masked language modeling,” *arXiv preprint arXiv:2401.01099*, 2024.
- [17] M. Kim, M. Jeong, J. Y. Lee, and N. S. Kim, “Seginr: Segment-wise implicit neural representation for sequence alignment in neural text-to-speech,” *IEEE Signal Processing Letters*, vol. 32, pp. 646–650, 2025.
- [18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, J. Burstein, C. Doran, and T. Solorio, Eds. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186. [Online]. Available: <https://aclanthology.org/N19-1423/>
- [19] V. Sitzmann, J. Martel, A. Bergman, D. Lindell, and G. Wetzstein, “Implicit neural representations with periodic activation functions,” in *Advances in Neural Information Processing Systems*, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds., vol. 33. Curran Associates, Inc., 2020, pp. 7462–7473. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2020/file/53c04118df112c13a8c34b38343b9c10-Paper.pdf
- [20] B. Desplanques, J. Thienpondt, and K. Demuynck, “ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification,” in *Proc. Interspeech 2020*, 2020, pp. 3830–3834.
- [21] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, “Vision mamba: Efficient visual representation learning with bidirectional state space model,” in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, and F. Berkenkamp, Eds., vol. 235. PMLR, 21–27 Jul 2024, pp. 62 429–62 442. [Online]. Available: <https://proceedings.mlr.press/v235/zhu24f.html>
- [22] H. Zen, V. Dang, R. Clark, Y. Zhang, R. J. Weiss, Y. Jia, Z. Chen, and Y. Wu, “Libritts: A corpus derived from librispeech for text-to-speech,” *Interspeech 2019*, 2019.
- [23] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *International Conference on Machine Learning*. PMLR, 2023, pp. 28 492–28 518.
- [24] T. Saeki, D. Xin, W. Nakata, T. Koriyama, S. Takamichi, and H. Saruwatari, “Utmos: Utokyo-sarulab system for voicemos challenge 2022,” in *Interspeech 2022*, 2022, pp. 4521–4525.
- [25] J. Kim, J. Kong, and J. Son, “Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 5530–5540.
- [26] Z. Zhang, L. Zhou, C. Wang, S. Chen, Y. Wu, S. Liu, Z. Chen, Y. Liu, H. Wang, J. Li *et al.*, “Speak foreign languages with your own voice: Cross-lingual neural codec language modeling,” *arXiv preprint arXiv:2303.03926*, 2023.